

EXTREME VALUE ANALYSIS

Purpose: To review the methods for estimating return periods for extreme hydrological events

I. EXTREME VALUE ANALYSIS: DEFINITIONS

Selection of the sample of extreme values

There are two approaches to the selection of the extremes which are to be fitted to an extreme value distribution:

1. Annual maximum series

All annual maxima are considered, i.e. one per year.

For statistical purposes, it is necessary to ensure the independence of the maxima (e.g. problem of a maximum in December followed by one in January) and therefore it is best to use the water year.

2. Partial duration series

Here, a threshold is selected and all the peaks over this threshold are considered for the N years of record. Again, care has to be exerted to avoid picking dependent peaks. FSR suggests that 'peaks should be separated in time by 3 times the time to peak and that the flow should decrease between peaks to two thirds of the first peak'.

These two methods will yield different results which should however be close for return periods of 10 or more years.

Frequency or Extreme Value Analysis

Definition: It is:

- (1) the estimation of the probability of exceedance of given levels of the variable in question, and/or
- (2) the estimation of the peak value which is likely to be *equalled or exceeded on average once in any specified period, T years*

Let us assume first that we are looking at annual maxima.

II. ANNUAL MAXIMA

Extreme value theory

Consider a sample of size n . If the variables Z_i are independent and identically distributed (iid), then the cdf of the maximum value would be given by:

$$\begin{aligned} P\{M_n \leq x\} &= P\{Z_1 \leq x\}P\{Z_2 \leq x\}...P\{Z_n \leq x\} \\ &= [F_Z(x)]^n \end{aligned}$$

When $n \rightarrow \infty$, $M_n \simeq X$ where X has one of following three types of distributions:

- the extreme value distribution type I (Gumbel),
- the extreme value distribution type II (Fréchet),
- the extreme value distribution type III (Weibull).

Type I: $F(x) = \exp\left(-e^{-\left(\frac{x-\beta}{\alpha}\right)}\right)$ for $-\infty < x < +\infty$

Type II: $F(x) = \begin{cases} \exp\left(-\left(\frac{x-\beta}{\alpha}\right)^{-\gamma}\right) & \text{for } x > \beta \\ 0 & \text{for } x \leq \beta \end{cases}$

Type III: $F(x) = \begin{cases} \exp\left(-\left(\frac{x-\beta}{\alpha}\right)^{\gamma}\right) & \text{for } x < \beta \\ 0 & \text{for } x \geq \beta \end{cases}$

The Gumbel distribution

The most commonly used distribution in hydrology is the Gumbel distribution:

$$F(x) = P(X \leq x) = \exp\left\{-\exp\left[-\left(\frac{x-\beta}{\alpha}\right)\right]\right\}$$

It has two parameters: location (β) and scale (α)

which can be written:

$$P(X \leq x) = \exp\{-\exp[-y]\} = P(Y \leq y)$$

with the **Gumbel Reduced Variate** $y = (x - \beta)/\alpha$

Random variable Y has the following statistics:

$$E(Y) = 0.5772$$

$$Var(y) = \pi^2/6$$

$$g_1(Y) = 1.1396$$

which yields, for X :

$$E(X) = \beta + 0.5772\alpha$$

$$Var(X) = \frac{\pi^2}{6}\alpha^2$$

$$g_1(X) = 1.1396$$

Note that X has **constant positive skewness**

Using the Gumbel distribution

Once a Gumbel distribution has been fitted to annual maxima, it can be used for either of the following two purposes:

- (1) Find the probability of exceedance of a given level
- (2) Find the level which is exceeded with a given probability

Rather than referring to the probability of exceedance, a **Return Period (T)** is generally used. So, these two purposes are:

- (1) Finding the return period for a given level:

$$T(x) = 1/P(X > x)$$

- (2) Finding the level of given return period:

$$x(T) \text{ such that } P(X > x(T)) = 1/T$$

For the Gumbel distribution:

$$(1) \quad T(x) = 1/(1 - F(x)) = \frac{1}{1 - \exp(-e^{-(x-\beta)/\alpha})} = \frac{1}{1 - \exp(-e^{-y})}$$

$$(2) \quad y(T) = -\ln[-\ln(1 - \frac{1}{T})] \\ \Rightarrow x(T) = \beta - \alpha \ln[-\ln(1 - \frac{1}{T})]$$

In hydrology, sometimes the following approximation is used:

$$y(T) \approx \ln[T - \frac{1}{2}]$$

(which works well if $T > 10$ years)

Presenting the results and showing goodness-of-fit

In hydrology, the goodness-of-fit of a fit to an extreme-value distribution is represented on a plot that compares population and sample quantiles.

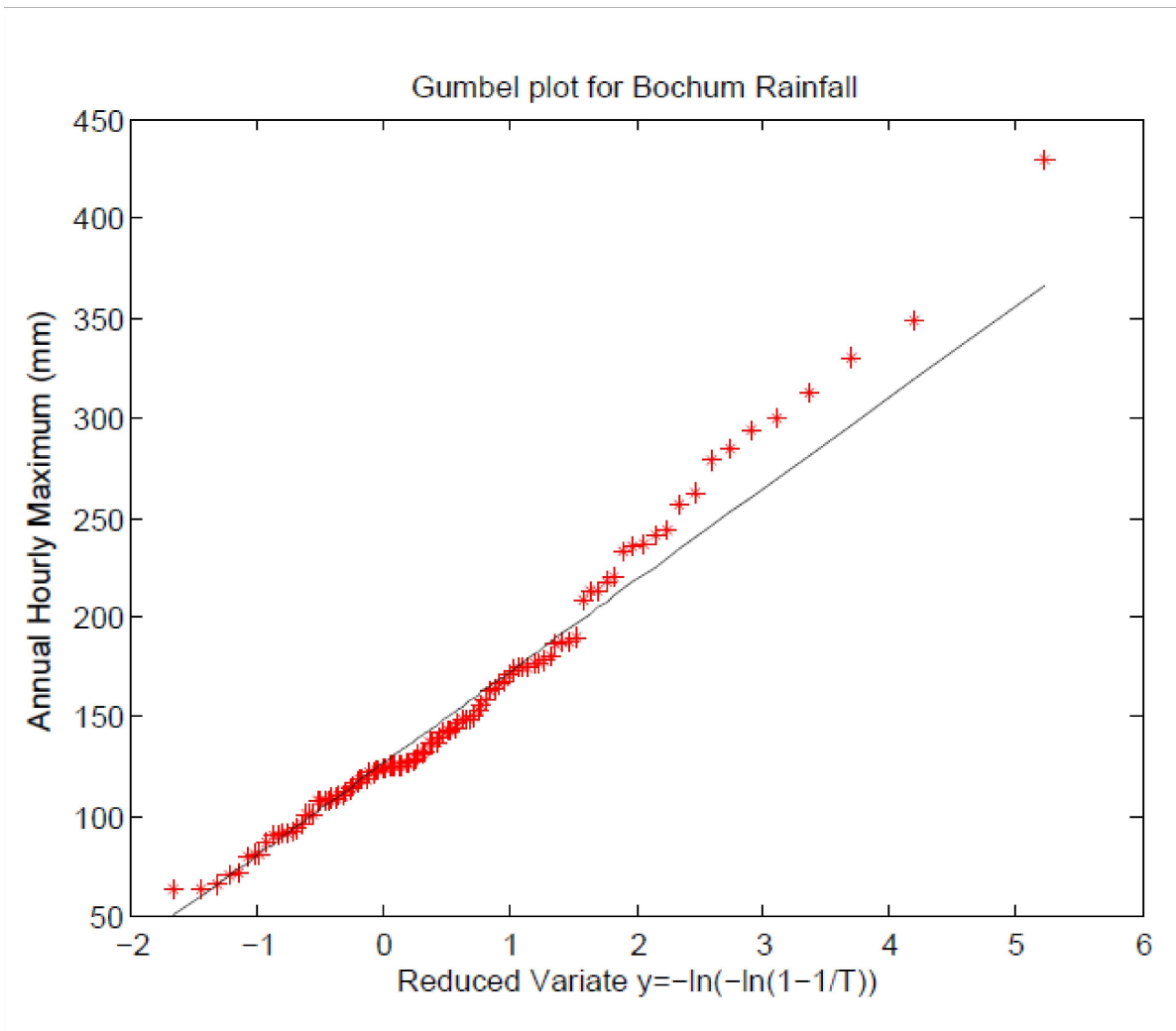
This is however not the usual q-q-plot. Rather, on the vertical axis, both quantiles are plotted, against the reduced variate on the horizontal axis. The latter is just a function of the probability of exceedance or the return period.

According to the Gumbel distribution the population quantile is:

$$x(T) = \beta + \alpha y(T) = \beta - \alpha \ln(-\ln(1 - \frac{1}{T}))$$

The sample quantile is obtained by what is called a **plotting position**, which is essentially a formula for the empirical cdf: the probability of exceedance of the m -th largest term in a sample of size N is generally estimated taken as $1/T = m/N$, but this is not a good formula for very skewed data like extremes. Others are preferred, e.g. the Gringorten formula:

$$T = (N + 0.12)/(m - 0.44)$$



In the figure above, a Gumbel distribution is fitted to the annual hourly maximum rainfalls at Bochum (Germany)

Fitting the Gumbel distribution

1. Method of moments

Given that we have sample estimators of $E(X)$ and $Var(X)$, i.e. the sample statistics $\bar{x}$ and $\hat{s}^2$, we can identify these to obtain:

$$\hat{\alpha} = \frac{\sqrt{6}\hat{s}}{\pi}$$

$$\hat{\beta} = \bar{x} - \frac{0.5772\sqrt{6}\hat{s}}{\pi}$$

2. Maximum likelihood method

The log-likelihood function is formed using the density function:

$$f(x) = d(P(X \leq x))/dx = \frac{e^{-(x-\beta)/\alpha}}{\alpha} \exp\{-e^{-(x-\beta)/\alpha}\}$$

as:

$$L(\alpha, \beta) = - \sum_{i=1}^N (x_i - \beta)/\alpha - \sum_{i=1}^N \exp[-(x_i - \beta)/\alpha] - N \ln \alpha$$

and by setting:

$$\partial L / \partial \alpha = 0 \text{ and } \partial L / \partial \beta = 0$$

we obtain estimates of α and β .

This has to be solved numerically

Limitations

1. The limiting distribution only applies properly if $n \approx 10^9$ whereas often, $n = 365$.
2. There is dependence between the values of Z_i of which the maximum is taken. In practice, Watson showed that the method is valid if the process is of the moving-average type, but there are possible problems with cycles and trends. Note that in hydrology, the station-year method consists in increasing the amount of data used for ev analyses by using several (neighbouring) gauges.
3. The Z_i are in practice not identically distributed because of seasonal variability.
4. There are a few negative numbers generated by a Gumbel distribution which has no lower bound.
5. The sampling errors of the parameter estimates are very large.

Generalised Extreme Value distribution

If the Gumbel distribution turns out to show a poor fit, it may be because either the Ev-Type II or the EV-Type III distribution is more appropriate.

There is a more flexible distribution that can therefore be used, namely the Generalised Extreme Value (GEV) distribution. This is defined by the equation:

$$\begin{aligned} P(X \leq x) &= \exp\left(-\left(1 + \xi \frac{x-\beta}{\alpha}\right)^{-1/\xi}\right) & \text{if } \xi \neq 0 \\ &= \exp\left\{-\exp\left[-\left(\frac{x-\beta}{\alpha}\right)\right]\right\} & \text{if } \xi = 0 \end{aligned}$$

This is defined,

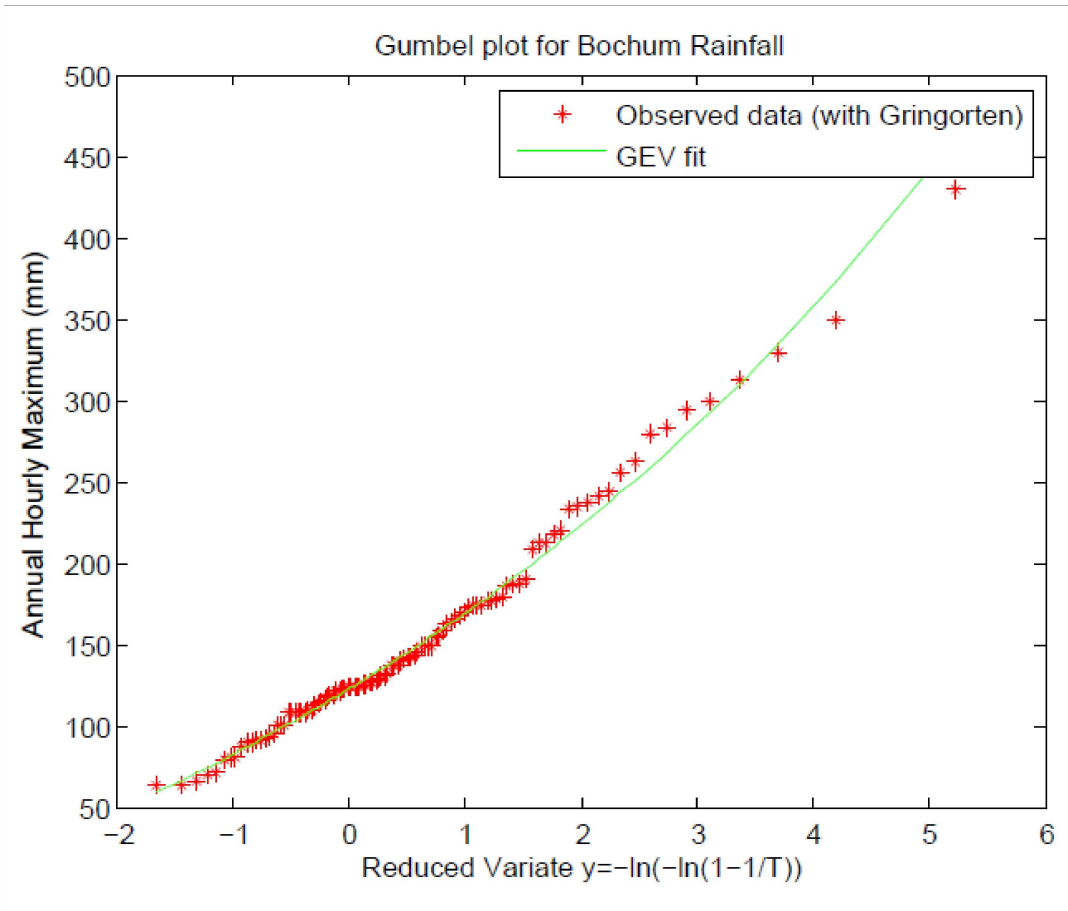
- . if $\xi > 0$ for $x > \beta - \alpha/\xi$
- . if $\xi < 0$ for $x < \beta - \alpha/\xi$
- . if $\xi = 0$ for $-\infty < x < +\infty$

In the first case, it is an EV-Type II (Fréchet) distribution, in the second, and EV-Type III (Weibull) distribution, and in the third, a Gumbel distribution.

To represent the above distribution on a Gumbel plot, the quantile $x(T)$ of the GEV distribution can be related to the quantile $y(T) = -\ln(-\ln(1 - 1/T))$ of the Gumbel distribution by the relation:

$$\boxed{x(T) = \beta + \alpha \{ \exp[\xi y(T)] - 1 \} / \xi}$$

where α , β and ξ are parameters of scale, location and shape respectively.



In the figure above, a GEV distribution is fitted to the annual hourly maximum rainfalls at Bochum (Germany)

Other distributions commonly used for hydrological extremes

Although there are no theoretical grounds for this, other distributions are often used aside from the extreme-value distributions. The following two are typical.

1. Log-normal distribution

This distribution is important not only for extreme values but more generally because it enables positively skewed variables to be seen as transforms of a Gaussian variable.

X is log-normally distributed if and only if $\ln(X - a)$ is normally distributed.

2. Pearson type III distribution

It is a gamma distribution with a third parameter, the location. It thus has a probability density function given by:

$$f(x) = \frac{(x-a)^{\alpha-1} \exp(-(x-a)/\beta)}{\beta^{\alpha} \Gamma(\alpha)}$$

and the parameters are easily estimated using the maximum likelihood method.

Note that the Log-Pearson distribution is also used, this being a distribution such the log of the variable is Pearson distributed.

III. PEAKS OVER THRESHOLD

Outline of the method

This method looks at the *Partial Duration Series* and the number and magnitude of the flows above a threshold L such that there are typically between 3 and 5 peaks per year. Unlike the use of annual maxima, we have a proper stochastic model representing not only intensities, but also inter-arrival times.

- The times between peaks u_i can be considered to be iid (independent and identically distributed) and a simple assumption is that they are exponentially distributed. This defines the process of arrivals of peaks as a *Poisson process*.

This is equivalent to saying that if on average there are Λ peaks per year which exceed the thresholds, then the number M of POT events per year is a random variable which has the following Poisson probability distribution:

$$P(M = m) = \Lambda^m e^{-\Lambda} / m!$$

- The magnitudes X of the peaks which exceed the threshold L can be assumed to be exponentially distributed with scale parameter Ω and location parameter L :

$$P(X > x | X > L) = \exp(-\Omega(x - L))$$

More generally, a Generalised Pareto Distribution with scale parameter Ω , location parameter L and shape parameter ξ can be used for the distribution of exceedances of the threshold:

$$\begin{aligned} P(X > x | X > L) &= \left(1 + \xi \Omega (x - L)\right)^{-1/\xi} && \text{if } \xi \neq 0 \\ &= \exp(-\Omega(x - L)) && \text{if } \xi = 0 \end{aligned}$$

This is defined,

- . if $\xi > 0$ for $x \geq L$
- . if $\xi < 0$ for $L \leq x \leq L - 1/(\xi \Omega)$
- . if $\xi = 0$ for $L < x < +\infty$

Using the POT model

Here, we focus upon the exponential case which is easier to fit.

- If we are looking for a level $x(T)$ corresponding to a return period of T years, we must note that this is a level that is exceeded on average once every ΛT events in the partial duration series, so that we can write the equation:

$$\exp(-\Omega(x(T) - L)) = 1/\Lambda T$$

This leads to the following expression for this T year event:

$$\boxed{x(T) = L + (\ln(\Lambda) + \ln(T))/\Omega}$$

- Conversely, if we seek to find the return period of events that equal or exceed a certain threshold x , i.e. their return period $T(x)$, then the same equation as above:

$$\exp(-\Omega(x - L)) = 1/\Lambda T(x)$$

must now be solved for $T(x)$, so that the return period is given by the following equation:

$$\boxed{T(x) = \exp(\Omega(x - L))/\Lambda}$$

Parameter estimation

Λ is estimated directly from the mean number of events per year.

For the estimation of Ω and of L , consider the following:

a- using the exponential distribution of the peak over the threshold L and taking the mean, we obtain:

$$E(X) = L + 1/\Omega$$

b- considering the ordered samples of exceedances of L (as random variables):

$X_{(1)}, \dots, X_{(N)}$ from the smallest to the largest

then the distribution of the smallest peak is easy to obtain:

$$\begin{aligned} P(X_{(1)} > x) &= [\exp(-\Omega(x(T) - L))]^N \\ &= \exp(-N\Omega(x(T) - L)) \end{aligned}$$

So, taking the means:

$$E(X_{(1)}) = L + 1/(N\Omega)$$

This provides us with two equations linking population statistics to the model parameters.

Therefore, by applying the *Method of Moments*, i.e. by identifying the sample estimates of these two statistics, $\bar{x}$ and $x_{(1)}$ with the population statistics $E(X)$ and $E[X_{(1)}]$ respectively, we obtain the following estimates of the other two parameters

$$\hat{\Omega} = (N - 1) / \{N(\bar{x} - x_{(1)})\}$$

&

$$L = x_{(1)} - 1 / (N\hat{\Omega})$$

The POT method is useful for the estimation of the magnitude and frequency of events which have a low return period T.

It has been shown however that for $T > 10$ years, the levels $x(T)$ calculated using the annual maximum series and the POT methods differ little.

More specifically, we have the following equivalences:

<i>Annual Maxima distribution</i>	<i>Peak-Over-Threshold exceedance distribution</i>
Gumbel	Exponential
GEV with shape parameter ξ	Generalised Pareto with shape parameter ξ